

School of Computer Science and Engineering
Department of Computer Science and Applications

Academic Year 2025-26

Mid Term Examination
Sept-October 2025

BCA20040– Data Structures Using C

Program	Bachelor of Computer Applications-BCA-Science	Semester	III
Specialization	-----	Duration	1 Hour
		Max. Marks	30

Instructions to the Candidate:

1. Write the PRN on the top right-hand corner of the question paper.
2. Draw neat diagrams.
3. Assume suitable data, if necessary.
4. Section 1 is Compulsory.

Section I (Compulsory to Solve all Questions)				
Questions	Marks	CO	Blooms Taxonomy	Unit no
a) Explain the difference between an algorithm and a program. State how Primitive data structures are different from Non Primitive data structures with their types and description.	5 Marks	1	Remembering	Unit- I
1 b) Explain the following data structures with diagram in brief 1. Arrays 2. Singly Linked List 3. Doubly Linked list 4. Circular Linked list 5. Double Pointers	5 Marks	2	Understanding	Unit-II

Section II
(Answer any 2 Questions)

2	How the doubly linked list is different from Circular linked list. Explain the structure of a doubly linked list. Write an C program to insert a node at the beginning of a Singly linked list.	10 Marks	2	Apply	Unit II
3	Write a program or pseudocode to implement a stack using Arrays. Explain how the push and pop operations work in your implementation by taking examples. Give the output for the above program	10 Marks	3	Analyse	Unit III
4	Explain the queue data structure in detail. Write a program to insert 5 elements into Queue and remove 2 elements from queue using an array implementation. Display the step by step output.	10 Marks	3	Apply	UNIT III

Best of Luck

Term End Examination December 2025

BCA20040 - Data Structures Using C

Question Paper ID: 063957

Faculty/School	School of Computer Science and Engineering	Term	Semester III
Program	BCA	Duration	1 hour 30 minutes
Specialization	-	Max. Marks	40

Section 1 - [7 Questions, 8 Marks] (5 X 8 Marks)
Answer any 5 questions

1	Discuss the importance of data structures in software development. And Explain the different operations that can be performed on data structures with examples :	8 marks	CO1	Understanding
2	Write a C program to create Circular Linked list and Delete a node from Circular linked list.	8 marks	CO2	Applying
3	Describe the process of converting an infix expression to a postfix expression using a stack. Provide a detailed algorithm and trace the conversion of a complex infix expression, such as $A + B * (C - D) / E$, showing the stack's state at each step.	8 marks	CO3	Evaluating
4	Write a C program to create a Binary Search Tree (BST) by inserting nodes. Then display the tree using Inorder Traversal. Your program should allow the user to enter n nodes and display the elements in sorted order.	8 marks	CO4	Analysing
5	Explain Heap Tree and its types. Implement MIN heap tree using Array	8 marks	CO4	Applying
6	Write a C program to sort an array of numbers in ascending order using Bubble Sort. Compare the performance of Bubble Sort Algorithm with Insertion Sort.	8 marks	CO6	Creating
7	What is a Collision Resolution Technique (CRT)? Explain any two techniques.	8 marks	CO7	Understanding

END OF QUESTION PAPER



School of Computer Science and Engineering
Department of Computer Science and Applications

Academic Year 2025-26

Mid Term Examination
Sept-October 2025

BCA10050 – Linear Algebra

Program SYBCA (LINEAR ALGEBRA)

Semester III

Specialization

Duration 1 Hour

Max. Marks 30

Instructions to the Candidate:

1. Write the PRN on the top right-hand corner of the question paper.
2. Draw neat diagrams.
3. Assume suitable data, if necessary.
4. Section I is Compulsory.

Section I
(Compulsory to Solve all Questions)

Questions	Marks	CO	Blooms Taxonomy	Unit no
A) Find the Inverse of given matrix by adjoint method $\begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$	5 Marks	CO4	Apply	Unit- I
B) 1) Define with example :				
a) Scalar Matrix	2 Marks	CO1	Understanding	Unit-I
b) Null Matrix				

24°

2) If $\begin{bmatrix} x+y & x+2 \\ 2x-y & 16 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ 1 & 3y+1 \end{bmatrix}$, then find the value of $x+y$.

3 Marks

CO2

Knowledge

Unit 1

Section II (Answer any 2 Questions)

a) Prove that a map

4 Marks

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation where $T(x, y) = (y, x+1)$

b) Determine if the following vectors are linearly independent. Justify your answer:

3 marks

CO2, CO4

Understanding, apply

Unit 2, Unit 3

$$V_1 = (1, 1, 1),$$

$$V_2 = (1, 1, 0)$$

$$V_3 = (1, 0, 0)$$

c) Show that

3 marks

$W = \{(x, y, z), y=3x, z=5x\}$ is a subspace of vector space $\mathbb{R}^3$

Solve the following system of linear equations:

3

$$\begin{aligned} x+2y-3z &= -4, & 2x+3y+2z &= 2, \\ 3x-3y-4z &= 11 \end{aligned}$$

10 Marks

CO4

Apply

Unit 2

1) Check whether given matrix is singular or not. Justify your answer.

$$\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 0 \\ -7 & 2 & 1 \end{bmatrix}$$

5 Marks

CO4

Apply

Unit 1

4

2) If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ then find

$$A^2 - 4A$$

5 Marks

CO4

Apply

Unit 1



Term End Examination December 2025

BCA10050 - Linear Algebra

Question Paper ID: 063954

Faculty/School	School of Computer Science and Engineering	Term	Semester III
Program	BCA	Duration	1 hour 30 minutes
Specialization	-	Max. Marks	40

Section 1 - [7 Questions, 8 Marks] (5 X 8 Marks)

Answer any 5 questions

1	Find the inverse of the following matrix if exist $A = \begin{bmatrix} 4 & 3 & 3 \\ -1 & 0 & -1 \\ -4 & -4 & -3 \end{bmatrix}$	8 marks	CO4	Evaluating
2	If $A = \begin{bmatrix} 5 & 4 \\ -2 & 3 \end{bmatrix}$ $B = \begin{bmatrix} -1 & 3 \\ 4 & -1 \end{bmatrix}$ then find C^T such that $3A - 2B + C = I$ where I is the identity matrix	8 marks	CO4	Evaluating
3	Find matrix P that diagonalizes given matrix A and determine $P^{-1}AP$ $A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$	8 marks	CO3	Applying
4	Solve the following system of equations : $\begin{aligned} x+y+z+w &= 6 \\ 2x+3y-w &= 0 \\ -3x+4y+z+2w &= 4 \\ x+2y-z+w &= 0 \end{aligned}$	8 marks	CO2	Evaluating
5	Write the vector $V = (2, 1, 5)$ as a linear combination of the vectors $U_1 = (1, 2, 1)$, $U_2 = (1, 0, 2)$ and $U_3 = (1, 1, 0)$	8 marks	CO2	Evaluating

6	A) Determine whether set of vectors $\{ (5, 5, 5), (4, 4, 0), (3, 0, 0) \}$ forms a basis for vector space R^3. Justify your answer. (4 marks) B) Is the following set of vectors Orthogonal? Justify your answer. (4 marks) $u_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \quad u_2 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} \quad u_3 = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$	8 marks	CO3	Applying
7	Given a map $T : R^3 \rightarrow R^3$ such that $T(x, y, z) = (x, x+y, x+y+z)$. Determine whether T is a linear transformation	8 marks	CO1	Applying

END OF QUESTION PAPER

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School of Computer Science and Engineering
Department of Computer Science and Applications

Academic Year 2025-26

Mid Term Examination
March 2026
BCA10060- Python

Program	BCA(Science)	Semester	IV
Specialization	-	Duration	1 Hour
		Max. Marks	30

Instructions to the Candidate:

1. Write the PRN on the top right-hand corner of the question paper.
2. Draw neat diagrams.
3. Assume suitable data, if necessary.
4. Section 1 is Compulsory.

Section I
(Compulsory to Solve all Questions)

Questions	Marks	CO	Blooms Taxonomy	Unit no
1 a) Differentiate between primitive data types and collection data types in Python with suitable examples.	5 Marks	1,3,4	Understanding	Unit- I
b) Explain the range() function with suitable examples.	5 Marks	1,3,4,7	Analysing	Unit-II

Section II
(Answer any 2 Questions)

2 a) Describe the various categories of operators available in Python. Illustrate each with examples.	10 Marks	1,3,4	Understanding	Unit -I
b) Explain the methods used to add elements to a list in Python with one example each				

3	a) What are mutable and immutable data types? Is a set mutable? Justify your answer. b) Explain the structure and features of the Dictionary data type in Python with suitable examples.	10 Marks	1,3,4,5	Analysing	Unit-III
4	a) Write a program using Python function that accepts a collection of numbers (use tuple) as an argument and returns the maximum and minimum values. b) Write a Python program to create a tuple containing the names of all 12 months of the year. Display the tuple of months. Then, using tuple slicing, display the months quarter-wise as eg:-First Quarter: January, February, March	10 Marks	1,3,4,5	Applying	Unit-III,IV

Best of Luck



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PRN: 1272240541

Term End Examinations May-June 2026.

BCA10060/BCA2PF03A/COS2120B - Python

Question Paper ID: 074763

Faculty/School	School of Computer Science & Engineering	Term	Sem IV
Program	SY_BCA	Duration	1 hour 30 minutes
Specialization		Max. Marks	40

Section 1 - [7 Questions, 8 Marks] (5 X 8 Marks)

Answer any 5 questions

1	a) Explain Python keywords and indentation with examples b) Perform list operations: append, remove, sort with example.	8 marks	CO1	Understanding
2	a) Explain tuple and its properties with example. b) Write a Python program to print Fibonacci series using a loop.	8 marks	CO1, CO2	Applying
3	a) Explain dictionary methods with example. b) Write a list comprehension to generate squares of numbers from 1 to 10.	8 marks	CO2, CO3	Evaluating
4	a) Demonstrate use of *args and **kwargs. b) Demonstrate set operations (union, intersection).	8 marks	CO2, CO3	Applying
5	a) Explain any two mathematical and two statistical functions in NumPy with example. b) Explain class variable and instance variable with example.	8 marks	CO2, CO3	Understanding
6	a) Create a DataFrame using a dictionary and demonstrate how to access rows and columns with an example. b) What is vectorization in NumPy explain it with example.	8 marks	CO3, CO4	Applying
7	Explain the concept of filtering in Pandas. Illustrate your answer with a suitable example.	8 marks	CO4	Evaluating

END OF QUESTION PAPER

School of Computer Science and Engineering
Department of Computer Science and Applications

Academic Year 2025-26

Mid Term Examination
March 2026

BCA20060-Computer Networks

Program	BCA (Science)	Semester	IV
Specialization	-	Duration	1 Hour
		Max. Marks	30

Instructions to the Candidate:

1. Write the PRN on the top right-hand corner of the question paper.
2. Draw neat diagrams.
3. Assume suitable data, if necessary.
4. Section 1 is Compulsory.

Section I
(Compulsory to Solve all Questions)

Questions	Marks	CO	Blooms Taxonomy	Unit no
a) Explain the different types of Transmission Modes.	5 Marks	CO1	Understand	Unit- I
1 b) Compare the TCP and UDP protocols of Transport Layer.	5 Marks	CO2	Understand	Unit-II

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Section II
(Answer any 2 Questions)

2	Explain the types of Network based on the Geographical area.	10 Marks	CO1	Understand	Unit- I
3	Explain the TCP/IP reference model with neat diagram	10 Marks	CO2	Understand	Unit- II
4	Draw the given Line Coding Scheme for data : 11101011 i) NRZ-I ii) NRZ-L iii) Manchester iv) Differential Manchester	10 Marks	CO3	Apply	Unit- III

Best of Luck

Approved
06/03/2026



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PRN: 1272240541

Term End Examinations May-June 2026.**BCA20060/COS2076B/COS2112B - Computer Networks**

Question Paper ID: 074762

Faculty/School	School of Computer Science & Engineering	Term	Sem IV/III
Program	SY_BCA	Duration	1 hour 30 minutes
Specialization	—	Max. Marks	40

Section 1 - [7 Questions, 8 Marks] (5 X 8 Marks)**Answer any 5 questions**

1	Explain the goals and applications of computer networks. Describe different types of network topologies with suitable diagrams.	8 marks	CO1	Remembering
2	Compare the OSI reference model and TCP/IP model in detail. Explain the functions of each layer in both models.	8 marks	CO2	Applying
3	(a) Explain digital data transmission concepts. (b) Encode the binary data 101011 using NRZ-I encoding. Show signal transitions.	8 marks	CO3	Applying
4	(a) Explain flow control mechanisms: Sliding Window and stop-wait. (b) Suppose the data word that is to be sent is: 1011 0011 1011 1011 0101 1010. Explain the checksum generator at the sender side and the checksum checker at the receiver side for the provided data.	8 marks	CO4	Remembering
5	Explain random access protocols such as ALOHA and CSMA/CD. Compare them with suitable examples.	8 marks	CO5	Remembering
6	(a) Explain IPv4 addressing and subnetting. (b) Given IP address 192.168.1.0/26, calculate: • Number of subnets • Number of hosts per subnet • Valid host range for the first subnet	8 marks	CO6	Applying
7	(a) Explain the features and differences between TCP and UDP. (b) Describe the working of DNS and the World Wide Web.	8 marks	CO7, CO8	Remembering

END OF QUESTION PAPER